

makes OpenLDAP such an exciting piece of software. OpenLDAP is very modular and can be thought of as having a frontend and a backend. The frontend basically is about the LDAP protocol, networking, etc, whereas the backend deals with the directory data storage.

At this point, it is sufficient to know that we can actually store data in an SQL database (MySQL, MS-SQL) and access that data through OpenLDAP using the 'sql' backend.

We can also create a proxy LDAP server, actually getting LDAP data from other servers using the 'meta' backend.

Normally, we use the Berkeley Data Base (bdb) backend with OpenLDAP servers, by default. The reasons and benefits for doing so can be explored in a subsequent article in this series.

It is this availability of backends and the possibility of using various backends together in a single instance of the OpenLDAP server that necessitates a prescribed structure for *slapd.conf*.

The default structure for *slapd.conf* as described in the man page (*man 5 slapd.conf*) is:

```
# comment - these options apply to every database
<global configuration options>
...
.....
.....
# first database definition & configuration options
database <backend 1 type>
..... <configuration options specific to backend 1>
..... <configuration options specific to backend 1>
..... <configuration options specific to backend 1>
# subsequent database definitions & configuration options
```

Global configuration options are defined at the top of *slapd.conf*. These options apply to all backends, unless explicitly overridden within a backend definition.

Databases are then defined by the keyword 'database'. Then database-specific options such as suffix, rootdn, rootpw, indexes, etc, are defined.

As we can see now, since we have only one database backend, specifying rootpw at the bottom of the file worked, as the rootpw is a database configuration option, but since it is not a global configuration option, specifying rootpw as the topmost line, failed.

Since *slapd.conf* requires a structured configuration, a utility called *slaptest* is included in the *openldap-servers* package. *slaptest* reads the configuration file and checks whether it is okay or not. We use the *-f* option to perform a sanity check on a configuration file:

```
[root@vbg ~]# slaptest -f /etc/openldap/slapd.conf
bdb_db_open: warning - No DB_CONFIG file found in directory /var/lib/ldap:
(2)
Expect poor performance for suffix dc=knaf,dc=org.
config file testing succeeded
[root@vbg ~]#
```

Note: *slaptest* is a very useful and important tool, especially in production environments, where configuration changes and updates must be thoroughly tested before applying them.

Access control restrictions

Since this article focuses more on securing the OpenLDAP server, we go back to our discussion on access control restrictions. ACLs (access controls) in *slapd.conf* in their most simple form can be described by the following syntax:

```
access to [what] by [who] [Permission]
```

As an example, the default access control restriction (when none is explicitly specified in the configuration file) is:

```
access to * by * read
```

By evaluating this syntax, it becomes:

- [What] => * (all parts of the directory)
- [Who] => * (all users - including anonymous)
- [Permission] => read

Thus the default access control restriction in *slapd.conf*, in plain English, means: "Allow all users to read all portions of the directory."

Can we allow anonymous writes into the directory as well? Let us test this by changing the access control restriction to: "Allow all users to write to all parts of the directory."

```
access to * by * write
```

Add this ACL to *slapd.conf* and restart the *ldap* service.

Let us also create a new address book entry, by the cn (common name) of 'lly user'. A copy of the sample LDIF used in the first article of this series is given below:

```
[vbg@vbg-work ~]$ cat /tmp/lly-user.ldif
dn:cn=lly user,ou=addressbook,dc=knaf,dc=org
cn: lly user
sn: user
l: Gurgaon
street: N-37, Old DF Colony, Sector-14
st: Maryana
postalCode: 122003
homePhone: 8124 3383383
mobile: 0955655555
mail: info@fosteringlinux.com
objectClass: top
objectClass: inetOrgPerson
```

Let us now try to add this entry to the directory:

```
[vbg@vbg-work ~]$ ldapadd -x -h 192.168.122.1 -f /tmp/lly-user.ldif
adding new entry "cn=lly user,ou=addressbook,dc=knaf,dc=org"
ldap_add: Strong(er) authentication required (8)
additional info: modifications require authentication
```